SUPPORTING ADULT GUIDANCE

Angles in a triangle

Angles Tell Us Why · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Turning Triangles (a version of the curriculum lead's own dynamic-geometry activity), learners drag the corners of a triangle and press Turn the triangles: two copies turn half a turn about the midpoints of CA and CB, carrying angles a and b to C, where they sit either side of c on a straight line. This works for any triangle the learner makes. The corners snap so that every angle is a whole number of degrees, so the three labels always add up to exactly 180°; equal sides show tick marks and right angles a square. Six because puzzles follow, then a stretch that explains the turning argument with a labelled diagram. Learners then find third angles, judge which triangles are possible, and make triangles to order.

Driving Question: *Does every triangle hide a straight line?*

The mathematics, in plain terms

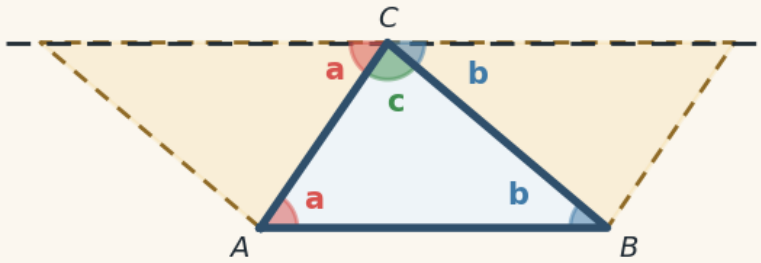
The angle sum of a triangle is 180°. The half-turn argument used here is a rotation proof: a half turn about the midpoint of a side swaps its endpoints and maps every line to a parallel line, so the copies' outer edges at C lie on one straight line parallel to AB. (The usual textbook proof uses alternate angles with a parallel line, which comes later.) Isosceles base angles are equal; equilateral triangles have three 60° angles; the two acute angles of a right-angled triangle sum to 90°.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Why 180°?

Why do the angles of this triangle add up to 180°?

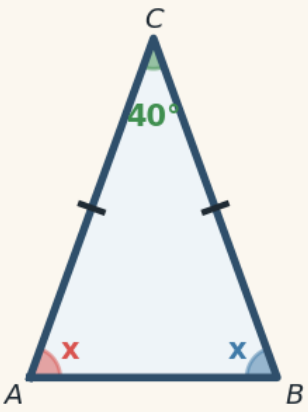


1. Turn a copy half a turn about the middle of CA.	angle a lands at C
2. Turn a copy half a turn about the middle of CB.	angle b lands at C
3. a, c and b now sit side by side on a straight line through C.	a + b + c = 180°

The turning works for every triangle, so every triangle's angles add up to 180°.

Example 2 — An isosceles triangle

The marked sides are equal. Find x.



1. The base angles are equal, so both are	x
2. Angles in a triangle add up to 180°.	$x + x + 40° = 180°$
3. So the two base angles share	$180° - 40° = 140°$
4. Each base angle is	$140° \div 2 = 70°$

x = 70°, because base angles of an isosceles triangle are equal and angles in a triangle add up to 180°.

See it for any triangle, then use it, then test what is possible

Learners reshape the triangle and turn it many times before the fact is stated, so 180° is something they have seen hold for every shape they tried. The stretch then explains why it must hold for all triangles. The special triangles and the “impossible triangle” statements apply the fact.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Turning Triangles and six because puzzles. Stretch: why every triangle's angles add up to 180°.	~16 min
Movement	The third angle: eight quick ones.	~7 min
Reflection	True or not? Which triangles can exist.	~7 min
Creativity	Make a triangle to order.	~10 min
Rest	Every triangle, the same.	~2 min

Nutrition	Intellectual: certainty.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking bigger triangles have a bigger angle sum.
Ask: “Make a tiny triangle and a big one. Do their corners still make a straight line?”

Taking the angle between the equal sides as a base angle.
Ask: “Which side is not marked? The base angles are at its ends.”

Forgetting the right angle when finding the third angle.
Ask: “Have you used all three angles, including the right angle?”

Halving the wrong amount in an isosceles triangle.
Ask: “What do the two equal angles share between them?”

Believing a triangle can have two obtuse angles.
Ask: “What would two angles over 90° add up to?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: angles on a straight line (Lesson 2). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says an isosceles triangle with a 100° angle has base angles of 100° too. The aim is a question that lets them test the total.

- “If two angles were 100°, what would they add up to?”
- “What must all three add up to?”

- “So can the 100° angle be a base angle?”
- “What is left for the two base angles?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why turning triangles, not tearing corners?
Both show the same thing. The turning version is dynamic, so learners can test any triangle, and it explains why the corners fit.

Why do the corners jump a little when dragged?
They snap so that every angle is a whole number of degrees. That keeps the three labels adding up to exactly 180°, with no rounding to explain.

Is the stretch required?
No. The secure core is using the 180° fact and the isosceles, equilateral and right-angled facts with reasons.

What comes next?
Lesson 4: because–therefore with quadrilaterals.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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